

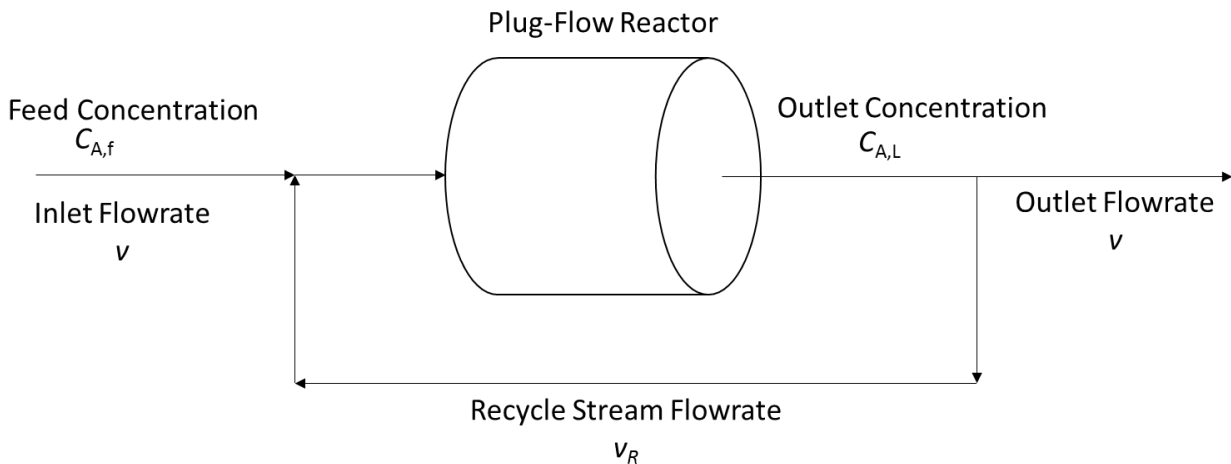
CHE 321 – Computer Exam

(Open Notes, Open Books)

Time: 8:00 – 9:45 am

Name: _____

Question: (100 points) Consider a plug-flow reactor with recycle as shown in Figure below.



A liquid phase reaction $2A \rightarrow B \rightarrow C$ is conducted in a plug-flow reactor. The rate expressions are given as $r_A = k_1 C_A^2$ and $r_B = k_2 C_B$. The feed contains only reactant A of concentration $C_{A,f}$ and it is mixed with a recycle stream containing unconverted reactant of concentration $C_{A,L}$ before entering the plug-flow reactor. The flowrates of feed stream is v and the flowrate of recycle stream is given as v_R . The plug-flow reactor has length L , cross-sectional area A , and volume V .

The physical parameters are given below

$$k_1 = 0.001 \text{ L mol}^{-1} \text{ s}^{-1}$$

$$k_2 = 0.0001 \text{ s}^{-1}$$

$$C_{A,f} = 40 \text{ mol L}^{-1}$$

$$L = 10 \text{ m}$$

$$A = 0.01 \text{ m}^2$$

$$v = 1 \text{ L s}^{-1}$$

1. **(50 points)** The steady-state continuity equation for A, B, and C in the recycle plug-flow reactor can be written as

$$\begin{aligned}(\nu + \nu_R) \frac{dC_A}{dV} &= -k_1 C_A^2 \\(\nu + \nu_R) \frac{dC_B}{dV} &= k_1 C_A^2 - k_2 C_B \\(\nu + \nu_R) \frac{dC_C}{dV} &= k_2 C_B\end{aligned}$$

The boundary conditions are given as

$$\begin{aligned}C_A(V=0) &= \frac{\nu C_{A,f} + \nu_R C_A(V=AL)}{\nu + \nu_R} \\C_B(V=0) &= \frac{\nu_R C_B(V=AL)}{\nu + \nu_R} \\C_C(V=0) &= \frac{\nu_R C_C(V=AL)}{\nu + \nu_R}\end{aligned}$$

Solve above ODEs to obtain concentrations of A, B, and C as function of volume V of reactor when $\nu = \nu_R$.

[Hint: Assume $C_A(V=AL) = C_{A,f}$, $C_B(V=AL) = 0$ and $C_C(V=AL) = 0$ in the boundary conditions and solve the above ODEs. Update values of $C_A(V=AL)$, $C_B(V=AL)$, $C_C(V=AL)$ in the boundary conditions using the solution of ODEs and solve the ODEs again. Iterate until the values of $C_A(V=AL)$, $C_B(V=AL)$, $C_C(V=AL)$ are converged.]

2. **(50 points)** Obtain conversion of A, defined as $\frac{C_A(V=AL)}{C_{A,f}}$, for recycle flowrates of $\nu_R = 0, 0.1, 0.5, 1, 10, 20, 100 \text{ L s}^{-1}$. Plot the conversion vs. recycle flowrates.